DG Interns Hub

Week 2 Report

Network Traffic Analysis, Nmap and Windows Logging

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Programme	DG Interns Hub - SOC / Cyber Security
Week	2
Submission date	28-30 October 2026
Tools used	VirtualBox, Kali Linux, Ubuntu (Apache), Wireshark 4.6.6, Nmap 7.99, Windows 11 PowerShell, Event Viewer, TryHackMe (free)

1. Objective

To learn how to read packets and Windows event logs so that the endpoint and IDS work in Week 3 makes sense. This week covered the TCP handshake and flags, Wireshark filters, how Nmap scan types look on the wire, and the key Windows Security Event IDs.

2. Lab Setup

Machine	Role	IP address
Kali Linux (VirtualBox)	Attacker / analyst, runs Wireshark and Nmap	192.168.56.102 (eth1, host-only)
Ubuntu (VirtualBox)	Target web server, Apache 2.4.66	192.168.56.101 (enp0s8, host-only)
Windows 11 laptop (host)	Windows target for Nmap and event log tasks	192.168.56.1

Each VM has two adapters: Adapter 1 is NAT (internet for apt) and Adapter 2 is Host-only (private lab network 192.168.56.0/24). Connectivity was confirmed with a 4-packet ping from Kali to Ubuntu (0% loss). A VirtualBox shared folder (/media/sf_captures) was used to save PCAP and Nmap output straight into the submission folder.

3. Hands-on Task 1 - Capture Traffic

Apache was installed on Ubuntu (`sudo apt install apache2`). Wireshark captured on Kali eth1 while ping, curl and nslookup were run against Ubuntu.

Observation	What was seen	Evidence
ICMP echo	4 echo requests (192.168.56.102 to .101) and 4 echo replies, e.g. frames 444-453	Findings/01_icmp.png
TCP three-way handshake	[SYN] frame 462, [SYN, ACK] frame 463, [ACK] frame 464 on port 80	Findings/02_tcp_handshake.png
HTTP GET	GET / HTTP/1.1 from curl (frame 465), answered by HTTP/1.1 200 OK (text/html); session ended with [FIN, ACK] and a final [ACK]	Findings/03_http_get.png
DNS query	Standard query A google.com sent to 192.168.56.101; Ubuntu is not a DNS server, so ICMP port unreachable came back	Findings/04_dns.png

PCAP: Captures/Task1_capture.pcapng

4. Hands-on Task 2 - Nmap on Own VMs

Scan	Command	Result
Host discovery -sn	nmap -sn 192.168.56.0/24	4 hosts up in 10.02 s: .1 (Windows host), .100 (VirtualBox DHCP), .101 (Ubuntu), .102 (Kali)
Version -sV	nmap -sV 192.168.56.101	80/tcp open http, Apache httpd 2.4.66 (Ubuntu); 999 ports closed (reset)
OS -O	sudo nmap -O 192.168.56.101	OS fingerprint guess for the Ubuntu VM (see nmap_O.txt)
Aggressive -A	sudo nmap -A 192.168.56.101	OS, version, scripts and traceroute combined (see nmap_A.txt)
SYN scan -sS	sudo nmap -sS 192.168.56.101	Port 80 open; other probed ports closed (see nmap_sS.txt)
Windows host	sudo nmap -sS -sV 192.168.56.1	Scan of the Windows laptop (see nmap_windows.txt)

SYN scan on the wire. The -sS capture was filtered with `tcp.flags.syn==1 && tcp.flags.ack==0`. It shows a burst of [SYN] packets from 192.168.56.102 to 192.168.56.101, all from the same source port and each to a different destination port (25, 80, 587, 21, 443, 8080, 23, 139, 111, 3389, 110, 1723 ...), all with Seq=0 and arriving within a fraction of a second. Open ports answer SYN-ACK, closed ports answer RST. This one-source / many-ports / very-short-time pattern is what Snort will be tuned to detect in Week 3.

Output files: nmap_*.txt (-oN) and nmap_*.xml (-oX) in the Captures folder. PCAP: Captures/nmap_syn_scan.pcapng. Screenshots: Findings/05 to 10 and 11.

5. Hands-on Task 3 - Windows Events

A Windows 11 laptop was used (no Windows VM was available). PowerShell was run as Administrator to generate each event, then located in Event Viewer using Filter Current Log.

Event	How it was generated	Where / Screenshot
4625 failed logon	net use \\localhost\C$ /user:fakeuser WrongPass1 (repeated)	Security - 11_event_4625
4624 successful logon	net use with the labuser account; Logon Type 3 (network)	Security - 12_event_4624
4720 new user	net user labuser P@ssw0rd123! /add	Security - 13_event_4720
4732 added to Administrators	net localgroup Administrators labuser /add	Security - 14_event_4732
7045 service installed	sc.exe create DummySvc binPath= ... then sc.exe delete DummySvc	System - 15_event_7045
1102 log cleared	wevtutil cl Security (run last)	Security - 16_event_1102

6. Hands-on Task 4 - PowerShell Hunting

All 4625 events were pulled with Get-WinEvent and a hashtable filter, exported to CSV, and failures were counted per account:

```
$ev = Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625}
$ev | Select TimeCreated, Id, @{n='Account';e={$_.Properties[5].Value}} | Export-Csv
failed_logons_4625.csv -NoTypeInformation
Import-Csv failed_logons_4625.csv | Group-Object Account | Sort-Object Count -Descending |
Format-Table Count, Name
```

The account that failed most often (fakeuser) stands out against normal activity, which is how a brute-force or password-spray attempt would be spotted. Screenshot: Findings/17_powershell_4625.png; CSV: Captures/failed_logons_4625.csv.

7. TryHackMe and Wireshark Samples

Completed rooms (each with its own writeup in the Writeups folder): Nmap, Live Host Discovery, Basic Port Scans, Windows Basics, Windows Fundamentals 1 and Windows Fundamentals 2, plus three official Wireshark sample captures.

8. Key Learnings

Topic	Takeaway
TCP	A normal connection is SYN, SYN-ACK, ACK, then FIN/ACK to close. Abnormal flag patterns point to scans.
Wireshark	Display filters make one story visible in thousands of packets; capture filters limit what is saved.
Nmap	-sn finds hosts, -sS is a fast half-open scan, -sV identifies services, -O guesses the OS, -A does everything and is the noisiest.
Windows logs	4625 + 4624 + 4720 + 4732 in sequence tells a compromise story; 1102 tells you someone is hiding it.
Baseline	Knowing what normal looks like (usual logons, usual protocols) is what makes anomalies visible to a SOC analyst.

9. Conclusion

The lab produced real traffic and real log evidence for each concept. The SYN-scan pattern captured in Task 2 and the Windows Event IDs from Task 3 are the base for Week 3, where Snort will be used to detect the scan and endpoint logs will be correlated with network data.

10. Submission Structure

DG Interns HUB / Week 2 / Notes, Writeups, Findings, Captures, Cheat Sheets, README.md, this PDF report and the presentation.